A Large-Scale Multi-Modal Benchmark and Robust Text-Free Inference for Breast Ultrasound Video Segmentation

Qiwei Wu
Shenzhen University
Shenzhen, China
wu_qi_wei@163.com

Zhihui Lai
Shenzhen University
Shenzhen, China
laizhihui@szu.edu.cn

Xiaoling Luo✉
Shenzhen University
Shenzhen, China
xiaolingluoo@outlook.com

Heng Kong
BaoAn Central Hospital of Shenzhen
Shenzhen, China
generaldoc@126.com

Foping Chen
Shenzhen University
Shenzhen, China
2100271078@email.szu.edu.cn

Junhong Zhang
Shenzhen University
Shenzhen, China
apple_zjh@163.com

Guangfei Liang
Shenzhen University
Shenzhen, China
gf_liang_study@163.com

Abstract

Breast cancer is one of the leading causes of death among women worldwide, and accurate segmentation of breast ultrasound images is of crucial importance. Compared with static images, videos have advantages in capturing lesions' dynamic and spatial information. However, research on breast tumor ultrasound video segmentation remains extremely limited due to the lack of high-quality publicly available datasets and the high cost of frame-by-frame annotation. In this situation, sparse-frame annotation of videos serves as a more affordable strategy, and diagnostic reports can be used to provide global semantic guidance for videos. However, diagnostic reports are unavailable in advance during clinical diagnosis, which limits the reasoning ability of current multi-modal models. To solve these issues, we construct the first large-scale multi-modal benchmark dataset, containing 1,085 videos with the corresponding clinical diagnostic text and tumor grading information. This dataset establishes a foundation for algorithm research and performance evaluation. Furthermore, we propose a multi-modal training framework with **T**ext-**F**ree inference for **B**reast tumor ultrasound **V**ideo **S**egmentation (TF-BVS). TF-BVS incorporates a **L**abel-aware **V**isual-**T**o-**T**ext **O**ptimal **T**ransport module (LV2TOT) to generate pseudo-text token embeddings highly consistent with clinical text distributions, thus enabling inference under text-free conditions in line with clinical workflows. TF-BVS employs a dual-branch joint training strategy, achieving robust segmentation performance even with less annotated training data. Extensive experiments demonstrate that TF-BVS achieves state-of-the-art segmentation results on both the proposed multi-modal breast ultrasound dataset and the public VTUS thyroid ultrasound dataset.

CCS Concepts

• **Computing methodologies** → **Video segmentation**.

Keywords

Multi-modal video segmentation, Breast tumor ultrasound video, Benchmark dataset, Text-Free inference

ACM Reference Format:
Qiwei Wu, Zhihui Lai, Xiaoling Luo, Heng Kong, Foping Chen, Junhong Zhang, and Guangfei Liang. 2026. A Large-Scale Multi-Modal Benchmark and Robust Text-Free Inference for Breast Ultrasound Video Segmentation. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26), November 10–14, 2026, Rio de Janeiro, Brazil.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3767308.3835065

1 Introduction

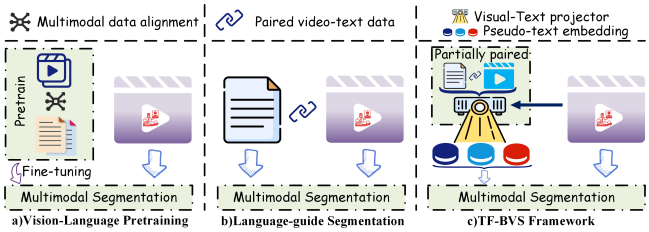


Figure 1: Comparison of multi-modal video segmentation paradigms. (a) VLP: Vision-Language Pretraining. (b) Language-guided Video Segmentation. (c) TF-BVS (ours): Text-Free inference Breast Ultrasound Video Segmentation.

Recent advances in breast tumor segmentation focus on 2D ultrasound images [10]. However, 2D images only capture the lesion at a single time and are easily affected by noise and artifacts, reducing segmentation accuracy. In contrast, ultrasound videos provide dynamic and spatial information as well as temporal continuity across frames, which can overcome noise in individual frames and improve the stability and accuracy of segmentation and diagnosis. Therefore, the methods based on ultrasound video have significant value for dynamic tracking and analysis of lesions [19].

Clinical diagnostic reports are often provided together with imaging data by the doctor, and there is a natural complementarity between images and text [24]. Textual information can effectively compensate for the limitations in quality of medical images [11, 14]. Vision-language pretraining (VLP) models [31] are pre-trained on

This work is licensed under a Creative Commons Attribution 4.0 International License.
MM '26, Rio de Janeiro, Brazil
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2213-4/2026/11
https://doi.org/10.1145/3767308.3835065

Table 1: Dataset statistics for existing breast tumor US video datasets and our dataset. Vid: number of videos. BF: number of annotated frames. BB: bbox. MA: masks. BM: benign or malignant. TE: clinical diagnostic reports. LE: tumor grade.

Dataset	Year	Vid	BF	BB	MA	BM	TE	LE
Li et al. [13]	2022	63	4619	×	✓	✓	×	×
Lin et al. [19]	2022	188	25272	✓	×	✓	×	×
Lin et al. [17]	2023	572	34300	✓	✓	✓	×	×
Ours	**2026**	**1085**	**50375**	✓	✓	✓	✓	✓

large-scale multi-modal data from various domains and then fine-tuned using unimodal video data, as shown in Figure 1(a), but they still lack domain-specific textual information embedding [37, 44]. Although some studies have explored text-guided video segmentation methods in natural scenes [2, 36], existing models still require text as input during inference, as shown in Figure 1(b). However, in real-world clinical diagnosis, doctors often write the diagnostic report after they have finished checking the ultrasound video. This means the report is not available beforehand, so current multi-modal methods cannot be used to assist with segmenting video frames in real time [41]. In addition, video annotation on a frame-by-frame basis is not only expensive and tedious, but contains a lot of redundant supervision information. In contrast, sparse-frame annotation assigns fine-grained labels only to a limited subset of frames at flexible intervals, which is more efficient. In addition, paired diagnostic text can further supplement lesion localization and morphological description.

Large-scale and diverse benchmarks are essential for advancing medical image analysis [23]. There are only unimodal breast tumor ultrasound video segmentation datasets, most of which are not publicly available and include limited videos and annotated frames [13, 17, 19]. These datasets lack multi-modal information, such as diagnostic reports and tumor grading information, limiting their clinical value and multi-modal modeling. To address these limitations, this paper builds the first large-scale multi-modal benchmark dataset for breast tumor ultrasound video segmentation, comprising 1,085 videos and 50,375 finely annotated frames, covering BI-RADS [8] grades from L2–L5 and providing paired diagnostic reports for each case. Our dataset significantly surpasses existing datasets in terms of scale and information diversity, leading to a solid foundation for further research. Detailed comparisons are provided in Table 1. Therefore, multi-modal breast tumor ultrasound video segmentation (MBTUVS) has three major challenges: **(1) lack of high-quality and publicly available multi-modal breast ultrasound video segmentation datasets with diagnosis text**, **(2) lack of text-free inference multi-modal methods**, **(3) lack of robust methods for sparse annotation data**.

To address the problems of textual reliance during inference and robustness in sparse annotation scenarios in MBTUVS, we propose the first multi-modal breast tumor ultrasound video segmentation method called **T**ext-**F**ree inference **B**reast tumor ultrasound **V**ideo **S**egmentation (TF-BVS), as shown in Figure 1(c). We design a novel **L**abel-aware **V**ision-**T**o-**T**ext **O**ptimal **T**ransport (LV2TOT) module, which uses only partial paired data to generate pseudo text token embeddings that match the distribution of real text. Efficient alignment between visual features and the textual semantic space is achieved under the guidance of tumor grade labels, which also reduces textual reliance during inference. In addition, to deal with sparse annotation scenarios, we design a joint training strategy that combines a cross-frame temporal backbone model and a frame-wise temporal branch model, achieving collaborative enhancement and effective temporal memory propagation between the branches. Our main contributions of this paper are as follows:

- We construct the first large-scale multi-modal breast tumor ultrasound video segmentation benchmark dataset, including paired diagnostic reports and tumor grading information, which provides a solid foundation for algorithm research and performance evaluation.
- The first multi-modal breast tumor ultrasound video segmentation model TF-BVS is proposed. It contains a novel tumor grades label-aware LV2TOT module. This module generates pseudo-text token embeddings that match the real-world text distribution, which helps to reduce textual reliance in real clinic diagnosis scenarios.
- A new joint training strategy is designed, which uses a cross-frame temporal backbone to guide a frame-wise temporal branch. It effectively propagates supervisory signals under sparse annotations and corrects bias in key-frame memory to suppress noise propagation.
- Extensive experiments on our self-constructed dataset and the public unimodal dataset VTUS demonstrate that our method outperforms mainstream approaches in segmentation performance. To the best of our knowledge, TF-BVS can be regarded as the first baseline method in the field of MBTUVS.

2 Related Work

2.1 Ultrasound video segmentation

In ultrasound video segmentation, early studies such as the multi-frame feature aggregation method [19] combined information from neighboring frames to reduce the sparsity caused by frame-by-frame processing so as to improve segmentation accuracy, but struggled with temporal consistency over long sequences. To address this problem, the spatio-temporal memory (STM) network [16] enhanced long-term temporal connections via memory banks, but it is resource intensive and sensitive to noise. BaS [28] used an efficient encoder-decoder for speed and accuracy, though it lacks adaptability to dynamic frame changes. FLA-Net [17] uses frequency domain information to gather temporal features and introduces localization-aware contrastive loss to improve consistency between frames, but this method is still limited in robustness for sparse annotation data. To better separate spatial and temporal features, DPSTT [13] proposed a dynamic memory mechanism, which helps handle irregular deformations and noise, but suffers from memory drift when annotations are sparse. [40] models long-term dependencies for non-rigid changes but requires high-quality annotations. For sparse annotation situations, MemSAM [7] utilizes temporal memory enhancement for stability. Although unimodal ultrasound video segmentation has made progress, the reliance on high-quality pixel-level annotations and challenges such as blurry boundaries, speckle noise, and non-rigid deformations in breast ultrasound still

cause temporal drift and error accumulation. To address the limitations of unimodal methods, this paper introduces clinical diagnostic text as complementary information to guide segmentation and designs a dual-branch collaborative training framework that works well in sparse annotation scenes. This framework makes full use of temporal information from unannotated frames and improves the robustness of the model in segmentation.

2.2 Text-guided Object Segmentation

Most multi-modal image and video segmentation methods build on vision-language pretraining (VLP) models [31], learning from paired text and visual data to enhance understanding and segmentation accuracy [22, 35, 37, 44]. However, medical images differ greatly from natural images, making it challenging for natural scene methods to capture the fine semantics required for tumor segmentation. Referformer [36] improves generalization with a language query framework but relies heavily on high-quality text. MUTR [39] designs a unified dual-modal framework to strengthen temporal consistency across frames, but has high inference cost. To solve the problems in cross-modal fusion and temporal modeling, SgMg [26] uses frequency-domain operations to enhance global cross-modal feature fusion, but it is easily affected by noise. Methods based on large vision-language models, such as VISA [38] and GLUS [18] improve reasoning under complex instructions, but fail to segment small targets and have an expensive training cost. ReferDINO [15] enhances dense prediction performance through location guidance, but still has the problem of textual reliance. Some static medical image segmentation methods try to relieve texture reliance by extracting target position information from text to generate pseudo labels or construct prototype spaces to approximate semantic guidance [41, 42]. However, they all rely on structured text and perform well in specific tasks, but poorly in scenarios like breast ultrasound, where reports' styles are diverse. Therefore, we propose a new paradigm, training a projection from visual space to text space with partially paired data, generating pseudo-text embeddings with a distribution close to real text to guide the segmentation on unimodal data and relieve text dependence.

3 Method

3.1 Overview

We propose TF-BVS, a multi-modal ultrasound video segmentation method for breast tumors, to reduce textual reliance during inference and to improve robustness under sparse annotation. As shown in Figure 2, the model uses a ReferDINO [15] encoder-decoder backbone with an adaptive memory update mechanism, which is supervised only on annotated key frames. To effectively reduce textual reliance, a tumor grades label-aware optimal transport module (LV2TOT) is designed to project visual features into a pseudo-text embedding space that matches the distribution of clinical reports. The entropy-regularized OT loss [6, 30] with a category-sensitive cost matrix penalizes grade mismatches. The backbone's predicted masks on key frames are used to initialize memory for the branch that segments unannotated frames. Learnable memory update and fusion mechanisms adaptively propagate temporal information, enhance key-frame memory, and improve overall robustness in scenarios with sparse annotation.

3.2 Cross-Frame Temporal Backbone

The interval between sparsely annotated frames can be flexibly set when constructing the dataset. In our dataset, we perform fine-grained annotations every five frames starting from frame 0, and denote the set of key-frame indices as $\mathcal{K} = \{0, 5, 10, 15, \ldots\}$. Existing methods have difficulty in dealing with changes in lesion shapes across frames in sparsely annotated medical videos, which usually cause errors in memory propagation. To address this problem, we designed a multi-modal cross-frame temporal backbone model based on the ReferDINO framework and designed a dynamic adaptive gating mechanism in the memory update process. This can improve temporal consistency and make the model robust to rapid changes in lesions.

For the t-th key frame of each video, the visual encoder extracts spatial features F_t, and the diagnostic report Q is encoded as a text feature E_Q. F_t together with E_Q are fed into the mask decoder to predict masks for key frames, and the supervision loss is computed only for $t \in \mathcal{K}$.

We design an adaptive gating mechanism on top of ReferDINO's memory-augmented object tracking. At each frame t, object embeddings O_t are aligned to the previous memory M_{t-1} using the Hungarian algorithm [15]:

$$\hat{O}_t = \text{Hungarian}(M_{t-1}, O_t). \tag{1}$$

Denoting the cosine similarity between aligned objects and memory as sim_t, we can compute its historical mean and variance via exponential moving averages (EMA):

$$\mu_t = (1-\lambda)\,\mu_{t-1} + \lambda\,\text{sim}_t, \tag{2}$$

$$\sigma_t^2 = (1-\lambda)\,\sigma_{t-1}^2 + \lambda\,(\text{sim}_t - \mu_t)^2. \tag{3}$$

With text-related gating, we normalize the similarity as

$$c_t' = \tfrac{1}{2}\big(1 + \cos(\hat{O}_t, f_t^{\text{cls}})\big) \in [0,1], \tag{4}$$

where f_t^{cls} is the [CLS] embedding of the text feature. The final adaptive update rate α_{eff} is given by

$$\alpha_\sigma = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) s_t \tag{5}$$

$$\alpha_{\text{eff}} = \alpha_\sigma \left[\gamma_{\text{gate}} + (1-\gamma_{\text{gate}}) c_t'\right], \tag{6}$$

where $s_t = \text{sigmoid}\,(\beta_\sigma(\sigma_t - \sigma_0))$, γ_{gate} is set as a learnable parameter and the memory is updated as:

$$M_t = (1-\alpha_{\text{eff}})\,M_{t-1} + \alpha_{\text{eff}}\,\hat{O}_t. \tag{7}$$

In our implementation, the hyperparameters are set as: $\lambda = 0.25$, $\alpha_{\min} = 0.05$, $\alpha_{\max} = 0.60$, $\beta_\sigma = 5.0$, $\sigma_0 = 0.05$. This design ensures high responsiveness to rapid changes. In addition, key-frame supervision provides reliable segmentation and robust memory guidance for the frame-wise branch.

3.3 Label-aware Visual-To-Text Optimal Transport Module

In breast tumor ultrasound diagnosis reports, the text style is diverse and the structure is often weak, making it difficult for traditional point-to-point distance methods to handle the complex data distribution. Optimal Transport (OT) [6] can achieve global alignment of feature distributions, which is more suitable for cross-modal semantic matching in this scenario. Based on this observation, we add tumor grade labels as a constraint in the OT framework to

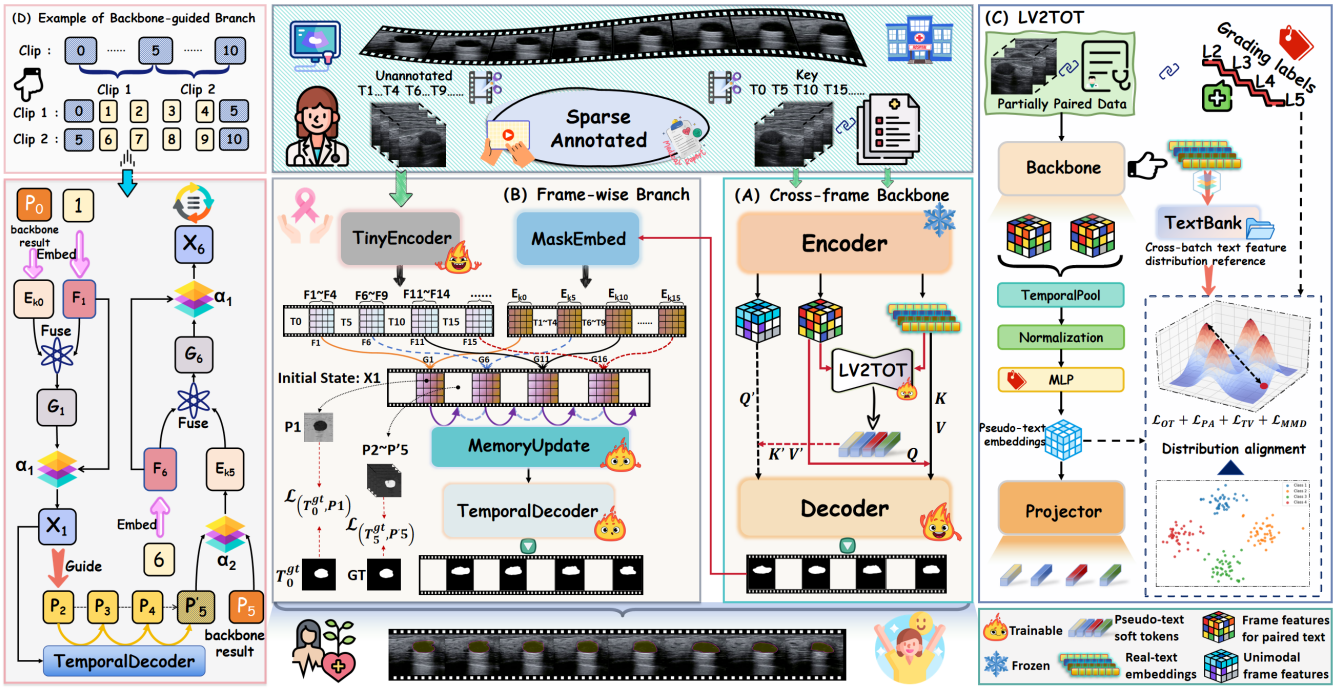


Figure 2: Overview of TF-BVS. The architecture comprises (A) a cross-frame temporal backbone, (B) a frame-wise temporal branch, and (C) the LV2TOT module structure. (D) illustrates an example of the backbone-guided branch for segmentation.

encourage pseudo-text and real-text features from cases with the same grade to match closely, while penalizing mismatches across different grades. Let $I \in \mathbb{R}^{B\times T\times C\times H\times W}$ denote a batch of input video frames, and $T_{gt} \in \mathbb{R}^{B\times L\times d}$ be the text embedding sequences from paired clinical reports, where L is the number of tokens. For each frame t of video i, the backbone extracts visual features $V_{i,t} \in \mathbb{R}^D$. The pseudo-text embedding $\hat{T}_{i,t} \in \mathbb{R}^d$ is then generated by a learnable MLP mapping f_θ:

$$\hat{T}_{i,t} = f_\theta\big(V_{i,t},\, g(i)\big), \quad t = 1, \ldots, T, \tag{8}$$

where $g(i) \in \{2, 3, 4, 5\}$ is the tumor grade label of video i. The global pseudo-text embedding $\bar{T}_i$ is obtained by temporal averaging $\bar{T}_i = \frac{1}{T}\sum_{t=1}^{T} \hat{T}_{i,t}$.

Label-aware OT and auxiliary losses. The real text embedding for video j, i.e. $E_{T,j} \in \mathbb{R}^d$, is defined as the [CLS] embedding from pooling T_{gt}. To enhance the robustness of cross-modal alignment, we introduce entropy-regularized OT between $\bar{T}_i$ and $E_{T,j}$. We collect the indices of annotated frames into set $\mathcal{S} = \{i \mid m_i = 1\}$, where $m_i \in \{0, 1\}$ indicates whether paired real text is available. The label-aware optimal transport cost matrix for $\mathcal{S}$ is defined as:

$$C_{ij} = \|\bar{T}_i - E_{T,j}\|_2^2 + \lambda_g\, \mathcal{I}_{g(i)\neq g(j)}, \quad i, j \in \mathcal{S}, \tag{9}$$

where $\mathcal{I}$ is the indicator of grade mismatch. Uniform marginals $a_\mathcal{S}, b_\mathcal{S} \in \mathbb{R}^{|\mathcal{S}|}$ are set as $a_\mathcal{S} = b_\mathcal{S} = [1/|\mathcal{S}|, \ldots, 1/|\mathcal{S}|]$. The entropy-regularized OT loss defined as:

$$\mathcal{L}_{\text{OT}} = \mathbf{1}_{|\mathcal{S}|>0} \cdot \min_{P\in U(a_\mathcal{S}, b_\mathcal{S})} \langle P, C\rangle - \varepsilon H(P), \tag{10}$$

where P is the transport plan, $H(P) = -\sum_{i,j} P_{i,j} \log P_{i,j}$ is the entropy of P, $\mathbf{1}_{|\mathcal{S}|>0}$ is an indicator function, and ε controls the entropy regularization strength. Pairwise alignment (PA) is computed when text exists:

$$\mathcal{L}_{\text{PA}} = \mathbf{1}_{|\mathcal{S}|>0} \cdot \frac{1}{|\mathcal{S}|} \sum_{i\in\mathcal{S}} \big(1 - \cos(\bar{T}_i, E_{T,i})\big). \tag{11}$$

The Maximum Mean Discrepancy (MMD) [9] aligns distributions in mixed batches. For videos with ground truth, real-text related terms are characterized by:

$$\begin{aligned}\mathcal{L}_{\text{MMD}} = &\frac{1}{B^2}\sum_{i,i'} k(\bar{T}_i, \bar{T}_{i'}) \\ &+ \mathbf{1}_{|\mathcal{S}|>0} \cdot \frac{1}{|\mathcal{S}|^2} \sum_{j,j'\in\mathcal{S}} k(E_{T,j}, E_{T,j'}) \\ &- \mathbf{1}_{|\mathcal{S}|>0} \cdot \frac{2}{B\,|\mathcal{S}|} \sum_{i=1}^{B}\sum_{j\in\mathcal{S}} k(\bar{T}_i, E_{T,j}),\end{aligned} \tag{12}$$

where $k(x, y) = \exp\left(-\frac{\|x-y\|_2^2}{2\sigma^2}\right)$ is the Gaussian kernel. To ensure temporal coherence of pseudo-text embeddings, we apply temporal total variation smoothness (TV) defined as follows to all videos:

$$\mathcal{L}_{\text{TV}} = \frac{1}{B(T-1)} \sum_{i=1}^{B}\sum_{t=1}^{T-1} \|\hat{T}_{i,t+1} - \hat{T}_{i,t}\|_1. \tag{13}$$

Finally, the combined loss is

$$\mathcal{L}_{\text{m}} = \lambda_1\, \mathcal{L}_{\text{OT}} + \lambda_2\, \mathcal{L}_{\text{PA}} + \lambda_3\, \mathcal{L}_{\text{MMD}} + \lambda_4\, \mathcal{L}_{\text{TV}}, \tag{14}$$

where the λ's are weighting coefficients, and the default value is set as $\lambda_1 = 5.0$, $\lambda_2 = 3.0$, $\lambda_3 = 2.0$, $\lambda_4 = 2.0$. This module can effectively reduce distribution mismatch and semantic ambiguity, thereby improving the effectiveness of pseudo-text. During inference, the input unimodal video is first classified by the backbone network to obtain the tumor grade label. Then, the LV2TOT projection generates pseudo-text that matches the label, which helps guide the decoder to predict the segmentation mask.

3.4 Frame-Wise Temporal Branch and Joint Training Strategy.

To fully exploit temporal information in unannotated frames within sparsely annotated videos, we design a key-frame-guided frame-wise temporal branch module.

Video slicing and feature extraction. The video is clipped by key frames, and each part contains two key frames and the unannotated frames in between. For each unannotated frame I_t, a lightweight convolutional encoder TinyEncoder$(\cdot)$ extracts features F_t.

Mask guidance and feature fusion. The starting key-frame mask M_{k_i} is embedded via MaskEmbed$(\cdot)$ denoted as E_{k_i} and fused with the first unannotated frame's features through a learnable weighted fusion operation Fuse$(\cdot,\cdot)$ to form the guidance feature:

$$G_t = \text{Fuse}\big(F_t,\ E_{k_i}\big). \tag{15}$$

The fused feature G_t is combined with image features through a learnable gating parameter α_1 to obtain the initial hidden state:

$$X_t = F_t + \alpha_1 \cdot G_t. \tag{16}$$

Temporal memory propagation and segmentation prediction. A convolutional Gated Recurrent Unit (GRU) recursively updates the hidden state Z_t using the input X_t and the previous hidden state Z_{t-1}, which, together with skip-connection features F_t, is fed into the decoder to produce the segmentation mask:

$$Z_t = \text{GRU}(X_t, Z_{t-1}),\ P_t = \text{TemporalDecoder}(Z_t, F_t). \tag{17}$$

At the last key frame of the segment, the prediction from the frame-wise branch $P'_{k_{i+1}}$ and the prediction from the cross-frame temporal backbone $P_{k_{i+1}}$ are fused via the gating parameter α_2, and the fused prediction is mapped into the state space to initialize the next segment:

$$Z_{\text{init}}^{(i+1)} = \alpha_2 \cdot P'_{k_{i+1}} + (1-\alpha_2)\cdot P_{k_{i+1}}. \tag{18}$$

At the inference stage, the outputs from both branches are fused at key frames using a gating parameter α_2. A visualization of this process is provided in Figure 2 (see "(D) Example of the backbone-guided branch for segmentation").

By dividing the video into clips by key frames, the model always has stable and reliable memory anchors during segmentation. This reduces drift and errors in long sequences. Key frames have high-quality labels, and the backbone model is fully supervised on these frames. Accurate target memory from key frames is passed to unlabeled frames, and then further spread using the frame-by-frame temporal relationship. This improves both the accuracy and consistency of sequence segmentation. At each key frame, the memory can be corrected by fusing the predictions from the branch model, making the key frame memory more stable. As a result, the model is more robust with sparsely labeled data.

Joint training strategy. We split the video into clips by key frames, and denote the starting and ending key frames of the i-th segment as k_i and k_{i+1}, respectively, with unlabeled frames $k_i{+}1, \ldots, k_{i+1}{-}1$ in between. We impose supervision only at two positions: (1) use the ground-truth mask of the starting key frame $P^{\text{gt}}_{k_i}$ to supervise the prediction of the first frame of the segment (i.e., frame $k_i{+}1$), (2) use the ground-truth mask of the ending key frame $P^{\text{gt}}_{k_{i+1}}$ to supervise the prediction of the ending key frame itself. At last, the overall loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{m}} + \sum_i \Big[\mathcal{L}\big(P^{\text{gt}}_{k_i}, \hat{P}_{k_i+1}\big) + \mathcal{L}\big(P^{\text{gt}}_{k_{i+1}}, \hat{P}_{k_{i+1}}\big)\Big], \tag{19}$$

where $\mathcal{L}(\cdot,\cdot)$ is the frame-wise segmentation loss, and no explicit supervision is applied to other unlabeled frames within the segment.

This joint training strategy makes full use of the high-quality labels from key frames to supervise both the backbone and the branch models. Because the tumor shape and echoes change very little between neighboring frames in ultrasound videos, there is strong continuity across frames. As a result, using nearest-neighbor supervision from $k_i \to k_i{+}1$ reliably passes the initial memory through unlabeled frames in each segment. This helps to prevent errors from building up and spreading.

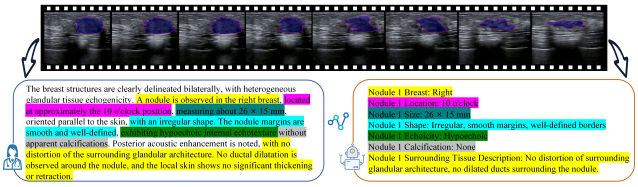


Figure 3: Example of using Lingshu32B [12] to perform text cleaning and structuring of original reports.

4 Experiment

4.1 Setup

Datasets. Due to the limited availability of multi-modal breast tumor ultrasound video segmentation datasets, we construct the first large-scale benchmark, comprising 1,085 videos with complete clinical reports and 302,746 RGB frames at a resolution of 256×256 pixels. Multiple experts sparsely annotate the videos at one mask every five frames, yielding 50,375 labeled frames. Each video is de-identified and assigned an L2-L5 grade; the dataset is split 6:2:2, with its class distribution reported in Table 3. We use Lingshu32B [12] to clean and structure the reports, as illustrated in Fig. 3. Experiments also use the public unimodal thyroid ultrasound video dataset VTUS, which contains 100 videos [40].

Metrics. We evaluate segmentation performance using the following metrics: region similarity $\mathcal{J}$ (Jaccard index), contour accuracy $\mathcal{F}$ (boundary F-measure), Dice coefficient, 95th percentile Hausdorff distance (HD95), and average surface distance (ASD).

Implementation. The Cross-frame temporal branch of TF-BVS strictly follows the configuration of ReferDINO and adopts a pre-trained GroundingDINO [21] as the backbone. Model training is performed on four RTX-4090 GPUs. TF-BVS contains 22.59M parameters and achieves an inference speed of 70.28 FPS.

Table 2: Segmentation performance of the proposed method and state-of-the-art methods on our self-constructed multi-modal dataset. TG: Whether to use text guidance for segmentation. Rate: Ratio of paired multi-modal data used in training. TFI: Whether text input is not required during inference.

Methods	TG	Rate	TFI	Venue & Year	$\mathcal{J}$(%)↑	$\mathcal{F}$(%)↑	Dice(%)↑	HD95(px)↓	ASD(px)↓
Unimodal									
LiVOS [20]	×	-	-	CVPR'25	62.71	57.82	70.20	22.36	9.21
Cutie [4]	×	-	-	CVPR'24	67.95	58.30	76.98	21.07	8.05
BaS [28]	×	-	-	JBHI'25	56.73	31.20	69.31	44.37	15.88
Vivim [40]	×	-	-	TCSVT'25	72.60	58.55	82.21	19.54	6.86
GDKVM [33]	×	-	-	ICCV'25	68.32	60.55	76.25	17.01	5.91
Multi-modal									
SAMWISE [5]	✓	100%	×	CVPR'25	71.02	63.31	78.44	17.33	6.68
GLUS [18]	✓	100%	×	CVPR'25	70.59	62.07	78.52	17.41	6.92
MUTR [39]	✓	100%	×	AAAI'24	61.90	53.98	71.41	29.32	12.64
VISA [38]	✓	100%	×	ECCV'24	65.07	56.46	74.20	26.14	10.48
ReferDINO [15]	✓	100%	×	ICCV'25	74.20	67.00	82.13	17.86	6.52
SgMg [26]	✓	100%	×	ICCV'23	74.25	64.79	82.14	17.59	6.58
Referformer [36]	✓	100%	×	CVPR'22	73.12	64.97	82.29	17.03	6.11
Losh [43]	✓	100%	×	CVPR'24	70.40	55.99	80.45	21.37	7.35
SOC [25]	✓	100%	×	NeurIPS'23	73.95	66.92	82.36	19.19	7.27
TF-BVS (Ours)	✓	**50%**	✓	-	**75.34**	**68.10**	**83.69**	**16.48**	**5.83**

Table 3: Number of video samples for each tumor grade in the multi-modal dataset.

Grade	Train	Valid	Test	All
L2	81	19	14	114
L3	413	142	146	701
L4	147	51	52	250
L5	10	5	5	20

4.2 Comparison With Other Methods

To the best of our knowledge, no open-source multi-modal model is specifically designed for breast tumor ultrasound video segmentation; therefore, we compare TF-BVS with representative unimodal and multi-modal video segmentation methods in Table 2. Most multi-modal methods are designed for natural scenes with clear boundaries and rich textures, whereas breast ultrasound videos contain substantial noise and blurred lesion boundaries, limiting their performance. Ultrasound-specific methods, including BaS [28], GDKVM [33], and Vivim [40], use only unimodal information and therefore depend heavily on high-quality annotations. On VTUS, we further evaluate different frame intervals to simulate sparse annotation settings, as shown in Table 4.

In contrast, TF-BVS employs adaptive memory updating to capture temporal changes under sparse annotations. LV2TOT requires partially paired data to generate pseudo-text embeddings, providing guidance without real text during inference. Moreover, the training strategy exploits unannotated frames and improves memory propagation between key frames, maintaining robustness under sparse supervision. As shown in Fig. 4, natural-scene methods often produce over-segmentation or under-segmentation, with errors accumulating over time because of noise and blurred boundaries. TF-BVS instead localizes tumor regions accurately and produces clearer boundaries, demonstrating stable memory propagation and consistent segmentation throughout the video sequence.

Table 4: Segmentation performance of the proposed method and state-of-the-art methods on the unimodal dataset VTUS. In 3: the frame interval of the training data is set to 3. In 5: the frame interval of the training data is set to 5.

Methods	Type	Data Rate	Dice(%)↑	$\mathcal{J}$(%)↑
Image				
UNet [32]	image	100%	66.62	53.28
UNet++ [47]	image	100%	76.56	64.86
TransUNet [3]	image	100%	74.61	62.50
SETR [45]	image	100%	72.88	60.10
DAF [34]	image	100%	77.16	65.83
Video				
OSVOS [29]	video	100%	77.69	67.54
ViViT [1]	video	100%	76.10	64.59
STM [27]	video	100%	78.98	68.97
AFB-URR [16]	video	100%	79.30	69.57
DPSTT [13]	video	100%	80.63	71.17
FLA-Net [17]	video	100%	80.42	70.75
RMem [46]	video	100%	78.04	67.75
MemSAM [7]	video	100%	79.22	71.01
Vivim [40]	video	100%	83.24	73.91
TF-BVS (Ours)	video	**100%**	**83.51**	**74.13**
TF-BVS (Ours)	video	**33.71%(In 3)**	83.25	73.67
TF-BVS (Ours)	video	**20.42%(In 5)**	81.48	71.05

4.3 Ablation Study

We evaluated the dual-branch architecture of TF-BVS, the impact of frame intervals during training and inference, and the effectiveness of the distribution alignment loss on our custom dataset. We visualized the pseudo-text distribution and the real text distribution generated by the LV2TOT module for different grades of tumor. Additionally, comprehensive ablation studies were conducted to verify the contribution of each core component and to compare our approach with existing text-free inference methods.

Temporal fusion with dual branches. Table 5 evaluates training intervals of 5 and 10 frames and the inference intervals of the cross-frame backbone (Te.M) and frame-wise branch (Te.B) using 50% paired data. With a 5-frame training interval, matching cross-frame inference outperforms frame-wise inference, demonstrating the

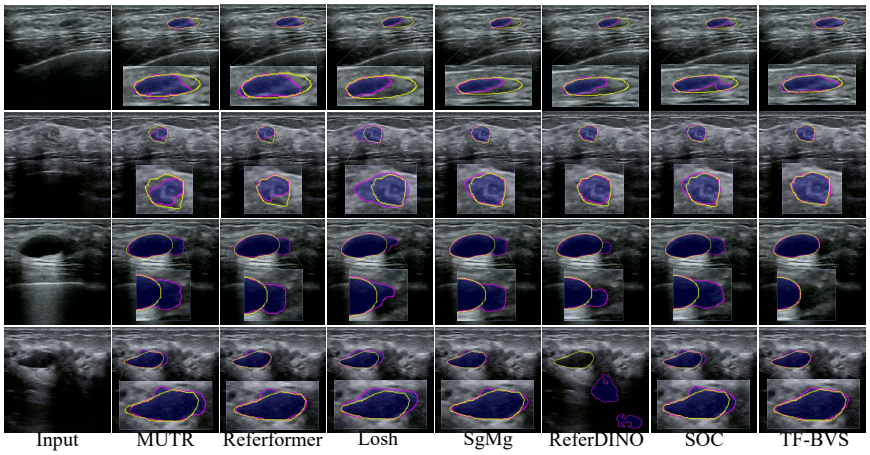
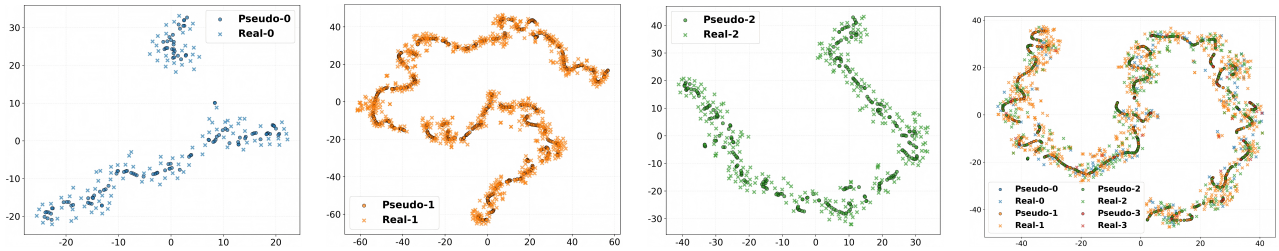


Figure 4: Visual comparison with state-of-the-art methods on the multi-modal dataset. The purple region represents the predicted area, while the yellow boundary denotes the ground truth lesion boundary.



(a) (b) (c) (d)

Figure 5: Comparison between the pseudo-text distribution and the real text distribution for different case categories: (a) L2; (b) L3; (c) L4; (d) overall distribution comparison.

Table 5: Ablation results for cross-frame backbone and frame-wise branch. Mo.: module used in training. M: cross-frame temporal backbone. B: frame-wise temporal branch. (Unit: % for $\mathcal{J}$, $\mathcal{F}$, and Dice; pixels for HD95 and ASD)

Mo.	Tr.	Te.M	Te.B	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	Dice↑	HD95↓	ASD↓
M	5	1	-	74.15	67.23	82.34	17.67	6.84
M	5	5	-	74.69	67.79	83.05	17.19	6.44
M+B	5	5	1	**75.34**	**68.10**	**83.69**	**16.48**	**5.83**
M	10	1	-	73.77	66.64	82.24	18.17	6.72
M+B	10	10	1	72.63	63.79	81.38	19.00	7.33

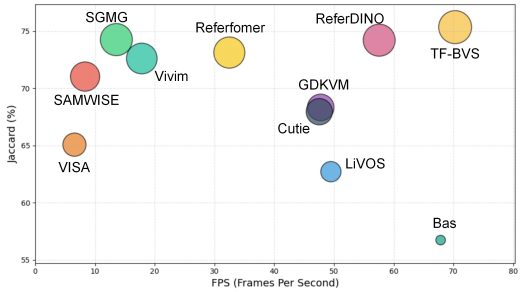


Figure 6: Comparison of FPS and Jaccard scores across different methods on the multi-modal dataset.

benefit of cross-frame temporal modeling. The frame-wise branch uses backbone-predicted key frames as memory anchors and adaptively fuses branch and backbone predictions at key frames. This bidirectional interaction corrects key-frame memory, suppresses error accumulation, and improves dense-frame segmentation under sparse supervision.

Text-free inference. Table 6 shows that LV2TOT generates effective pseudo-text guidance with only 50% paired data, while reducing the ratio to 40% causes only minor degradation. Fig. 5 further shows that the generated pseudo-text closely matches the real-text distribution within each category. Fig. 6 further compares the accuracy-efficiency trade-off across different methods.

Distribution alignment loss. We performed ablation experiments on four types of auxiliary distribution alignment losses for LV2TOT. All experiments use 50% paired multi-modal data and 50% unimodal data. During inference, predictions from the backbone and branch

Table 6: Comparison under different paired data ratios. TextR.: the proportion of paired-text data used for training. Tr.: training frame interval, Tr.F.: the number of training frames. (Unit: % for $\mathcal{J}$, $\mathcal{F}$, and Dice; pixels for HD95 and ASD)

TextR.	Tr.	Tr.F.	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	Dice↑	HD95↓	ASD↓
30%	5	30124	74.48	67.75	82.74	17.76	6.69
30%	10	15129	73.64	63.61	82.47	17.34	6.36
40%	5	30124	75.04	67.94	83.33	16.92	6.25
40%	10	15129	74.35	67.55	82.62	18.14	6.93
50%	5	30124	**75.34**	**68.10**	**83.69**	**16.48**	**5.83**
50%	10	15129	72.63	63.79	81.38	19.00	7.33
60%	5	30124	74.79	67.92	83.07	17.48	6.62
60%	10	15129	71.30	61.17	80.34	19.94	7.81

Table 7: Comparison of different combinations of distribution alignment losses. (Unit: % for $\mathcal{J}$, $\mathcal{F}$, and Dice; pixels for HD95 and ASD)

OT	PA	MMD	TV	$\mathcal{J}\uparrow$	$\mathcal{F}\uparrow$	Dice↑	HD95↓	ASD↓
✓				74.54	67.42	82.93	17.42	6.62
✓	✓			75.15	67.90	83.49	16.76	6.20
✓		✓		75.00	67.74	83.38	16.99	6.34
✓	✓	✓		75.24	67.92	83.64	16.56	6.04
✓	✓	✓	✓	**75.34**	**68.10**	**83.69**	**16.48**	**5.83**

are fused. The results (see Table 7) show that using these losses together helps the pseudo-text match the real text distribution more closely, and the TV loss further improves the temporal consistency of the pseudo-text.

Ablation of Model Components. Table 8 shows that removing text guidance (UnText) substantially degrades performance, confirming the importance of semantic constraints. Unlike the prototype-based Q-R baseline, LV2TOT uses grade-constrained Optimal Transport to generate informative pseudo-text embeddings. AD and Branch denote Dynamic Adaptive Gating and the dual-branch strategy, respectively, while Class-Emb shows that grade labels alone lack rich lesion semantics and are limited by classification accuracy. Overall, LV2TOT suppresses report noise while providing effective semantic guidance for segmentation.

Grade-label sensitivity. The grade conditions pseudo-text generation rather than directly determining the segmentation mask. As shown in Table 9, GT Grade achieves 75.77/84.03 in $\mathcal{J}$/Dice. Random-Wrong, which may retain adjacent-grade or within-risk-group semantics, causes only a modest drop to 75.02/83.14. In contrast, swapping labels between low-risk (L2/L3) and high-risk (L4/L5) groups reduces performance to 73.08/80.21, while removing the grade yields 73.16/81.48. These results show that segmentation is primarily visually driven, with grade semantics providing complementary guidance. Fig. 7 further shows the largest drop on L5 (7.08/7.51 in $\mathcal{J}$/Dice), indicating that high-risk cases are more sensitive to incorrect grade semantics.

5 Conclusion

This paper opens a new research direction on text-free multi-modal breast tumor ultrasound video segmentation, and we annotated the first large-scale multi-modal benchmark dataset. To address the limitations of existing methods in sparsely annotated scenarios and the textual reliance during inference, we proposed a multi-modal segmentation method called TF-BVS. An adaptive update mechanism for the backbone is developed to capture the dynamic changes of tumor regions across frames. We also designed a tumor grade label–aware visual-to-text optimal transport module (LV2TOT) that can generate effective pseudo-text embeddings to provide approximate semantic guidance to achieve text-free inference. In addition, a dual-branch joint training architecture is developed to synergistically improve both cross-frame and frame-wise temporal segmentation, making our model suitable for sparsely frame-annotated medical videos. Future work will focus on extracting tumor localization information from clinical diagnostic reports and transforming it into highly structured text to further enhance text guidance, as well as exploring more lightweight model designs.

Table 8: Ablation and comparison experiments.

Methods	TG	TFI	$\mathcal{J}(\%)\uparrow$	$\mathcal{F}(\%)\uparrow$	Dice(%)↑
Baseline	✓	×	74.20	67.00	82.13
+UnText	×	×	63.63	53.80	72.40
+Q-R	✓	✓	72.30	65.90	80.74
+V2TOT	✓	✓	73.16	66.89	81.48
+LV2TOT	✓	✓	73.96	67.59	82.25
+AD	✓	×	74.31	67.12	82.17
+AD+LV2TOT	✓	✓	74.15	67.23	82.34
+AD+Class-Emb	×	×	67.37	60.25	76.07
+AD+Branch	✓	×	75.30	67.96	83.57
+AD+Branch+LV2TOT	✓	✓	**75.34**	**68.10**	**83.69**

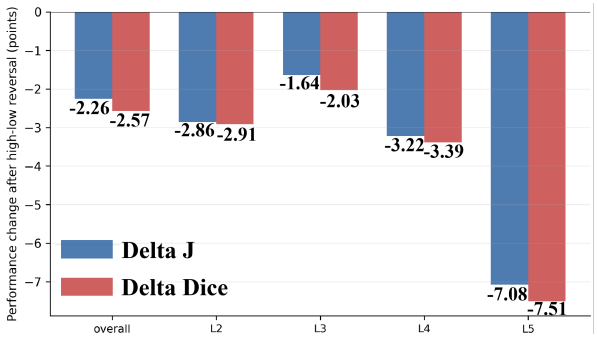


Figure 7: Grade-wise $\mathcal{J}$ and Dice drops under low/high-risk label swapping.

Table 9: Grade-label sensitivity analysis.

Metric	GT Grade	Random-Wrong	H/L Swap	w/o Grade
$\mathcal{J}$ (%)↑	**75.77**	75.02	73.08	73.16
Dice (%)↑	**84.03**	83.14	80.21	81.48

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant Nos. 62502320, 62476175, and 62272319, in part by the Natural Science Foundation of Guangdong Province under Grant Nos. 2026A1515050004 and 2023B1212060076, in part by the Shenzhen Science and Technology Program under Grant Nos. JCYJ20240813141424032 and JCYJ20250604145234045, and in part by the Scientific Foundation for Youth Scholars of Shenzhen University under Grant No. 827-0001083. This work was also supported by the Intelligent Computing Center of Shenzhen University.

References

[1] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. 2021. ViViT: A Video Vision Transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 6836–6846.

[2] Adam Botach, Evgenii Zheltonozhskii, and Chaim Baskin. 2022. End-to-End Referring Video Object Segmentation With Multimodal Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4985–4995.

[3] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L. Yuille, and Yuyin Zhou. 2021. TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation. arXiv preprint arXiv:2102.04306. doi:10.48550/arXiv.2102.04306

[4] Ho Kei Cheng, Seoung Wug Oh, Brian Price, Joon-Young Lee, and Alexander Schwing. 2024. Putting the Object Back into Video Object Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3151–3161.

[5] Claudia Cuttano, Gabriele Trivigno, Gabriele Rosi, Carlo Masone, and Giuseppe Averta. 2025. SAMWISE: Infusing Wisdom in SAM2 for Text-Driven Video Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3395–3405.

[6] Marco Cuturi. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. In *Advances in Neural Information Processing Systems*, Vol. 26. 2292–2300.

[7] Xiaolong Deng, Huisi Wu, Runhao Zeng, and Jing Qin. 2024. MemSAM: Taming Segment Anything Model for Echocardiography Video Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9622–9631.

[8] Carl J. D'Orsi, Edward A. Sickles, Ellen B. Mendelson, Elizabeth A. Morris, et al. 2013. *ACR BI-RADS Atlas: Breast Imaging Reporting and Data System* (5th ed.). American College of Radiology, Reston, VA.

[9] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. 2012. A Kernel Two-Sample Test. *Journal of Machine Learning Research* 13, 25 (2012), 723–773.

[10] Qinghua Huang, Yaozhong Luo, and Qiangzhi Zhang. 2017. Breast Ultrasound Image Segmentation: A Survey. *International Journal of Computer Assisted Radiology and Surgery* 12, 3 (2017), 493–507. doi:10.1007/s11548-016-1513-1

[11] Yu Huang, Chenzhuang Du, Zihui Xue, Xuanyao Chen, Hang Zhao, and Longbo Huang. 2021. What Makes Multi-Modal Learning Better than Single (Provably). In *Advances in Neural Information Processing Systems*, Vol. 34. 10944–10956.

[12] LASA Team, Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, Ruifeng Yuan, Jianyu Wang, Chenghao Xiao, Guizhen Chen, Chaoqun Liu, Zhaodonghui Li, Yu Sun, Junao Shen, Chaojun Wang, Jie Tan, Deli Zhao, Tingyang Xu, Hao Zhang, and Yu Rong. 2025. Lingshu: A Generalist Foundation Model for Unified Multimodal Medical Understanding and Reasoning. arXiv preprint arXiv:2506.07044. doi:10.48550/arXiv.2506.07044

[13] Jialu Li, Qingqing Zheng, Mingshuang Li, Ping Liu, Qiong Wang, Litao Sun, and Lei Zhu. 2022. Rethinking Breast Lesion Segmentation in Ultrasound: A New Video Dataset and a Baseline Network. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*. 391–400. doi:10.1007/978-3-031-16440-8_38

[14] Zihan Li, Yunxiang Li, Qingde Li, Puyang Wang, Dazhou Guo, Le Lu, Dakai Jin, You Zhang, and Qingqi Hong. 2024. LViT: Language Meets Vision Transformer in Medical Image Segmentation. *IEEE Transactions on Medical Imaging* 43, 1 (2024), 96–107. doi:10.1109/TMI.2023.3291719

[15] Tianming Liang, Kun-Yu Lin, Chaolei Tan, Jianguo Zhang, Wei-Shi Zheng, and Jian-Fang Hu. 2025. ReferDINO: Referring Video Object Segmentation with Visual Grounding Foundations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 20009–20019.

[16] Yongqing Liang, Xin Li, Navid Jafari, and Jim Chen. 2020. Video Object Segmentation with Adaptive Feature Bank and Uncertain-Region Refinement. In *Advances in Neural Information Processing Systems*, Vol. 33. 3430–3441.

[17] Junhao Lin, Qian Dai, Lei Zhu, Huazhu Fu, Qiong Wang, Weibin Li, Wenhao Rao, Xiaoyang Huang, and Liansheng Wang. 2023. Shifting More Attention to Breast Lesion Segmentation in Ultrasound Videos. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*. 497–507. doi:10.1007/978-3-031-43898-1_48

[18] Lang Lin, Xueyang Yu, Ziqi Pang, and Yu-Xiong Wang. 2025. GLUS: Global-Local Reasoning Unified into a Single Large Language Model for Video Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8658–8667.

[19] Zhi Lin, Junhao Lin, Lei Zhu, Huazhu Fu, Jing Qin, and Liansheng Wang. 2022. A New Dataset and a Baseline Model for Breast Lesion Detection in Ultrasound Videos. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*. 614–623. doi:10.1007/978-3-031-16437-8_59

[20] Qin Liu, Jianfeng Wang, Zhengyuan Yang, Linjie Li, Kevin Lin, Marc Niethammer, and Lijuan Wang. 2025. LiVOS: Light Video Object Segmentation with Gated Linear Matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8668–8678.

[21] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *Computer Vision – ECCV 2024*. 38–55. doi:10.1007/978-3-031-72970-6_3

[22] Timo Lüddecke and Alexander Ecker. 2022. Image Segmentation Using Text and Image Prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7086–7096.

[23] Xiaoling Luo, Qiaojian Zheng, Chengliang Liu, Yangneng Chen, Xiaoyan Dou, Tianyi Luo, Yingying Wen, Chao Huang, Yong Xu, and Jie Wen. 2026. A large-scale multi-view fundus images dataset and benchmark for diabetic retinopathy grading. *Information Fusion* 134 (2026), 104363. doi:10.1016/j.inffus.2026.104363

[24] Xiaoling Luo, Ruli Zheng, Qiaojian Zheng, Zibo Du, Shuo Yang, Meidan Ding, Qihao Xu, Chengliang Liu, and Linlin Shen. 2025. A Survey of Multimodal Ophthalmic Diagnostics: From Task-Specific Approaches to Foundational Models. arXiv preprint arXiv:2508.03734. doi:10.48550/arXiv.2508.03734

[25] Zhuoyan Luo, Yicheng Xiao, Yong Liu, Shuyan Li, Yitong Wang, Yansong Tang, Xiu Li, and Yujiu Yang. 2023. SOC: Semantic-Assisted Object Cluster for Referring Video Object Segmentation. In *Advances in Neural Information Processing Systems*, Vol. 36. 26425–26437.

[26] Bo Miao, Mohammed Bennamoun, Yongsheng Gao, and Ajmal Mian. 2023. Spectrum-Guided Multi-Granularity Referring Video Object Segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 920–930.

[27] Seoung Wug Oh, Joon-Young Lee, Ning Xu, and Seon Joo Kim. 2019. Video Object Segmentation Using Space-Time Memory Networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 9226–9235.

[28] Yan Pang, Yunhao Li, Teng Huang, Jiaming Liang, Ziyu Ding, Hao Chen, Baoliang Zhao, Ying Hu, Zheng Zhang, and Qiong Wang. 2025. Efficient Breast Lesion Segmentation from Ultrasound Videos Across Multiple Source-Limited Platforms. *IEEE Journal of Biomedical and Health Informatics* 29, 12 (2025), 8890–8903. doi:10.1109/JBHI.2025.3543435

[29] Federico Perazzi, Anna Khoreva, Rodrigo Benenson, Bernt Schiele, and Alexander Sorkine-Hornung. 2017. Learning Video Object Segmentation from Static Images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2663–2672.

[30] Gabriel Peyré and Marco Cuturi. 2019. Computational Optimal Transport: With Applications to Data Science. *Foundations and Trends in Machine Learning* 11, 5–6 (2019), 355–607. doi:10.1561/2200000073

[31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models from Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, 8748–8763.

[32] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. 234–241. doi:10.1007/978-3-319-24574-4_28

[33] Rui Wang, Yimu Sun, Jingxing Guo, Huisi Wu, and Jing Qin. 2025. GDKVM: Echocardiography Video Segmentation via Spatiotemporal Key-Value Memory with Gated Delta Rule. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 12191–12200.

[34] Yi Wang, Zijun Deng, Xiaowei Hu, Lei Zhu, Xin Yang, Xuemiao Xu, Pheng-Ann Heng, and Dong Ni. 2018. Deep Attentional Features for Prostate Segmentation in Ultrasound. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*. 523–530. doi:10.1007/978-3-030-00937-3_60

[35] Zhaoqing Wang, Yu Lu, Qiang Li, Xunqiang Tao, Yandong Guo, Mingming Gong, and Tongliang Liu. 2022. CRIS: CLIP-Driven Referring Image Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11686–11695.

[36] Jiannan Wu, Yi Jiang, Peize Sun, Zehuan Yuan, and Ping Luo. 2022. Language As Queries for Referring Video Object Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4974–4984.

[37] Hu Xu, Gargi Ghosh, Po-Yao Huang, Dmytro Okhonko, Armen Aghajanyan, Florian Metze, Luke Zettlemoyer, and Christoph Feichtenhofer. 2021. VideoCLIP: Contrastive Pre-Training for Zero-Shot Video-Text Understanding. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 6787–6800. doi:10.18653/v1/2021.emnlp-main.544

[38] Cilin Yan, Haochen Wang, Shilin Yan, Xiaolong Jiang, Yao Hu, Guoliang Kang, Weidi Xie, and Efstratios Gavves. 2024. VISA: Reasoning Video Object Segmentation via Large Language Models. In *Computer Vision – ECCV 2024*. 98–115.

[39] Shilin Yan, Renrui Zhang, Ziyu Guo, Wenchao Chen, Wei Zhang, Hongyang Li, Yu Qiao, Hao Dong, Zhongjiang He, and Peng Gao. 2024. Referred by Multi-Modality: A Unified Temporal Transformer for Video Object Segmentation. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 6 (2024), 6449–6457. doi:10.1609/aaai.v38i6.28465

[40] Yijun Yang, Zhaohu Xing, Lequan Yu, Huazhu Fu, Chunwang Huang, and Lei Zhu. 2025. Vivim: A Video Vision Mamba for Ultrasound Video Segmentation. *IEEE Transactions on Circuits and Systems for Video Technology* 35, 10 (2025), 10293–10304. doi:10.1109/TCSVT.2025.3563411

[41] Shuchang Ye, Mingyuan Meng, Mingjian Li, Dagan Feng, and Jinman Kim. 2024. Enabling Text-Free Inference in Language-Guided Segmentation of Chest X-Rays via Self-Guidance. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. 242–252. doi:10.1007/978-3-031-72111-3_23

[42] Shuchang Ye, Usman Naseem, Mingyuan Meng, and Jinman Kim. 2025. Alleviating Textual Reliance in Medical Language-Guided Segmentation via Prototype-Driven Semantic Approximation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 22316–22326.

[43] Linfeng Yuan, Miaojing Shi, Zijie Yue, and Qijun Chen. 2024. LoSh: Long-Short Text Joint Prediction Network for Referring Video Object Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14001–14010.

[44] Yan Zeng, Xinsong Zhang, Hang Li, Jiawei Wang, Jipeng Zhang, and Wangchunshu Zhou. 2024. X^2-VLM: All-in-One Pre-Trained Model for Vision-Language Tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 5 (2024), 3156–3168. doi:10.1109/TPAMI.2023.3339661

[45] Sixiao Zheng, Jiachen Lu, Hengshuang Zhao, Xiatian Zhu, Zekun Luo, Yabiao Wang, Yanwei Fu, Jianfeng Feng, Tao Xiang, Philip H. S. Torr, and Li Zhang. 2021. Rethinking Semantic Segmentation From a Sequence-to-Sequence Perspective With Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6881–6890.

[46] Junbao Zhou, Ziqi Pang, and Yu-Xiong Wang. 2024. RMem: Restricted Memory Banks Improve Video Object Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18602–18611.

[47] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. 2018. UNet++: A Nested U-Net Architecture for Medical Image Segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. 3–11. doi:10.1007/978-3-030-00889-5_1